Ömer Sabri Emeksiz

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Education

Koç University — Istanbul, Turkey
M.Sc. in Computer Science and Engineering, GPA: 3.92/4.00 — *October 2024 – September 2026*
Advisor: Prof. Metin Sezgin, Prof. Yücel Yemez, Prof. Engin Erzin

Research Focus: EEG-based cybersickness in immersive VR; Deep RL for affective computing on social robots

Marmara University — Istanbul, Turkey
B.Sc. in Electrical and Electronic Engineering, GPA: 3.33/4.00 (Honor, Ranked 5th) — *September 2021 – June 2024*
Advisor: Prof. Engin Maşazade

Research Focus: Reinforcement Learning

Experience

KUIS AI Center, Koç University — Istanbul, Turkey
AI Researcher, IUI Lab & MVGL Lab — *October 2024 – Present*

- Conducting research on EEG-based cybersickness detection and early prediction in immersive VR using time series modeling and computer vision. **Published at ICME 2026, SIU 2025.**
- Leading a 50-participant multimodal data collection study to establish the first open benchmark for cybersickness research, supported by European CHIST-ERA GENESIS (CHIST-ERA-20-BCI-003) and TÜBİTAK Project No. 221N367.
- Conducting research on deep reinforcement learning and affective computing for autonomous facial expression discovery on social robots. **Under review at a top ML venue.**
- Designed and implemented **Wrap-Up!**, an AI-powered collaborative meeting visualization tool: a full-stack Chrome extension for Google Meet that transcribes live audio (STT), orchestrates an LLM to extract meeting structure, and renders a shared real-time canvas with proactive AI facilitation and thread tracking.

Baykon Industrial Weighing Systems — Istanbul, Turkey
AI Engineer — *February 2024 – October 2024*

- Led and completed the TUBITAK–1501 industrial AI project, developing adaptive RL control policies (Monte Carlo, TD, Q-learning, MAB) to learn the coarse-to-fine feed switching point for industrial filling.
- Collected a 1,000-run filling dataset and deployed the learned policy on an embedded filling machine over Modbus/TCP, matching the operating point of the manually tuned PID baseline. **Published at SIU 2025, ASYU 2026.**

Marmara University — Istanbul, Turkey
Undergraduate AI Researcher — *December 2022 – June 2024*

- Conducted research on reinforcement learning for industrial filling process control, formulating the MDP framework and optimizing feed switching policy using Monte Carlo methods. **Published at ICECS 2023.**

TUBITAK — Istanbul, Turkey
Engineer Intern — *August 2023 – September 2023*

- Developed a continuous-time TCP/IP communication application using C# and a data extraction tool using Python.

Open Business Software Solutions (OBSS) — Istanbul, Turkey
Software Engineer Intern — *July 2023 – August 2023*

- Developed a full-stack enterprise-level Human Resources Job Postings and Applications Management System web application using Java, Spring Boot for backend and ReactJS, MUI for frontend.
- Used LDAP and LinkedIn API with Spring Security and JWT for authentication and authorization.
- Managed project with Apache Maven and utilized Hibernate, Spring Data JPA and MySQL for database.

Kalem Software — Istanbul, Turkey
Software Engineer Intern — *August 2022 – September 2022*

- Developed backend services for two B2B client web applications: File Organizer, a file storage and management platform, and Material Waybill Organizer, a system for materials organization, using Java, Spring Boot, Hibernate, and MySQL.

Publications

T. S. Jadidian, **Ö. S. Emeksiz**, E. Maşazade, S. Selim, "Uncertainty-Driven Deep Reinforcement Learning for Smart Industrial Filling", IEEE Innovations in Intelligent Systems and Applications Conference **(ASYU)**, 2026 **(accepted)**

Ö. S. Emeksiz, Y. Yemez, E. Erzin, T. M. Sezgin, "Early Prediction of Cybersickness from EEG in Immersive VR", IEEE International Conference on Multimedia and Expo **(ICME)**, 2026 **(accepted)**

Ö. S. Emeksiz, E. Erzin, Y. Yemez, T. M. Sezgin, "A Convolutional Transformer Model for EEG-Driven Cybersickness Detection in VR Experience", IEEE 33rd Signal Processing and Communications Applications Conference **(SIU)**, 2025

Ö. S. Emeksiz, E. Maşazade, S. Selim, "AI-Driven Optimization of the Filling Process: A Comparison of Reinforcement Learning Methods", IEEE 33rd Signal Processing and Communications Applications Conference **(SIU)**, 2025

Ö. S. Emeksiz, M. E. Ağcabay, E. Maşazade, S. Selim, T. Boysan, "A Reinforcement Learning Model for Industrial Filling Process Control", IEEE 30th International Conference on Electronics, Circuits and Systems **(ICECS)**, 2023

Professional Service

- **Reviewer:** IEEE ICME 2026, IEEE TMM, IEEE SIU 2026
- **Teaching Assistant:** Introduction to Computer Science and Programming; Computer Engineering Design; Communication Engineering

Research Interests

Reinforcement Learning · Time Series Modeling · Computer Vision · Affective Computing

Relevant Coursework

- Computer Vision
- Deep Learning
- Deep Unsupervised Learning
- Intelligent User Interfaces
- Medical Image Analysis
- Embodied Intelligence
- Biomedical Signal Processing
- AI in Health Sciences
- Image Processing

Skills

- **Programming Languages:** Python, C, C++, Java
- **Libraries and Frameworks:** PyTorch, NumPy, Scikit-learn, OpenCV, HuggingFace, Spring Boot, ReactJS